



Punjab University College
of
Information Technology
“PUCIT”

Chapter 06 – Application Software

GE-161 Introduction to Information and Communication
Technologies

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Learning Objectives

1. Describe what application software is, the different types of ownership rights, and the difference between installed and Web-based software.
2. Detail some concepts and commands that many software programs have in common.
3. Discuss word processing and explain what kinds of documents are created using this type of program.
4. Explain the purpose of spreadsheet software and the kinds of documents created using this type of program.

Learning Objectives

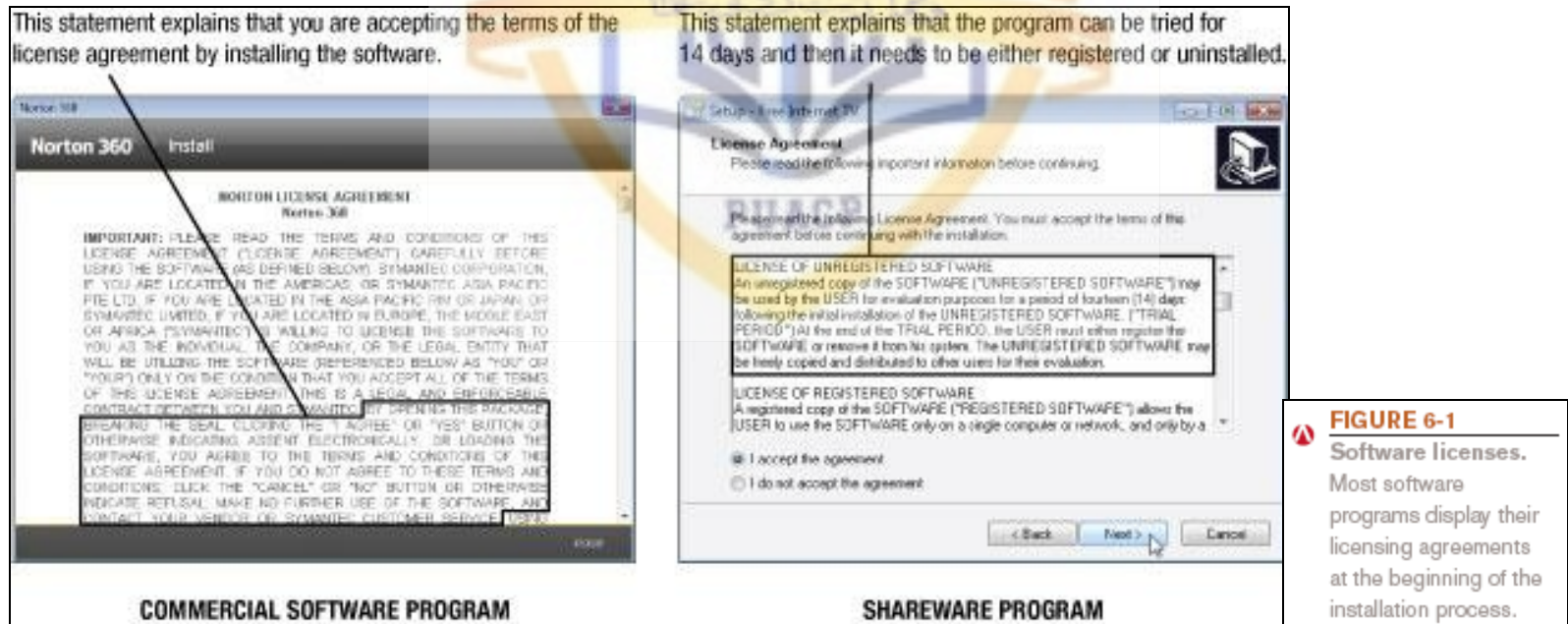
5. Identify some of the vocabulary used with database software and discuss the benefits of using this type of program.
6. Describe what presentation graphics and electronic slide shows are and when they might be used.
7. List some types of graphics and multimedia software consumers frequently use.
8. Name other types of application software programs and discuss what functions they perform.

Overview

- This chapter covers:
 - Characteristics of application software in general
 - A look at the most widely used types of application software, including:
 - Word processing
 - Spreadsheet
 - Database
 - Presentation graphics
 - Graphics and multimedia software
 - Other types of application software

The Basics of Application Software

- Software ownership rights: Specify the allowable use of the program
- Software license: Specifies the conditions under which a buyer of the program can use it



Software Ownership Rights

- Commercial software: Copyrighted software that is developed usually by a commercial company, for sale to others
- Shareware: Copyrighted software distributed on the honor system; consumers should either pay for it or uninstall it after the trial period
- Freeware: Copyrighted software programs that are given away by the author for others to use free of charge
- Public domain software: Software that is not copyrighted and ownership rights have been donated to the public domain
- Open source software: Programs with source code made available to the general public

Software Ownership Rights



FIGURE 6-2

Software ownership rights.

TYPE OF SOFTWARE	EXAMPLES
Commercial Software	Microsoft Office (office suite) Norton AntiVirus (antivirus program) Adobe Photoshop (image editing program) World of Warcraft (game)
Shareware	WinZip (file compression program) Ulead Video Toolbox (video editing/conversion program) Image Shrinker (image optimizer) Deluxe Ski Jump 3 (game)
Freeware	Internet Explorer (Web browser) OpenOffice.org (office suite) QuickTime Player (media player) Yahoo! Messenger (instant messaging program)
Public domain software	Lynx (text-based Web browser) Pine (e-mail program)

Desktop vs. Mobile Software

- Mobile phones and mobile devices typically require mobile software
 - Specifically designed for a specific type of device
 - Wide range of software available



Installed vs. Web-Based Software

- Installed software: Must be installed on the computer before it can be run
 - Can be purchased in physical form
 - Can be downloaded from the Internet



FIGURE 6-5
Installed software.
Installed software can be purchased in a physical package or downloaded via the Internet.

Installed vs. Web-Based Software

- Web-based software:
Run directly from the Internet
 - Also called Software as a Service (SaaS) and cloudware
 - Includes free software and fee-based software available via an application service provider (ASP)



Software Suites

- Software suite: Collection of software programs bundled together and sold as a single software package
 - Office suites are used by most businesses/individuals to produce documents
 - Typically include:
 - Word processing software
 - Spreadsheet software
 - Database software
 - Presentation graphics software
 - Additional productivity tools like calendars, messaging programs, or collaboration tools
 - Provides a common interface among programs in the suite

Software Suites

- Office suites include:
 - Microsoft Office
 - Apple iWork
 - OpenOffice.org (free)
- Most suites available in variety of versions
- Not all suites available for all OS
- Cost lower than buying each program separately



FIGURE 6-7
Office suites. Three of the most common commercial office suites are Microsoft Office, Corel WordPerfect Office, and Apple iWork.

Common Software Commands

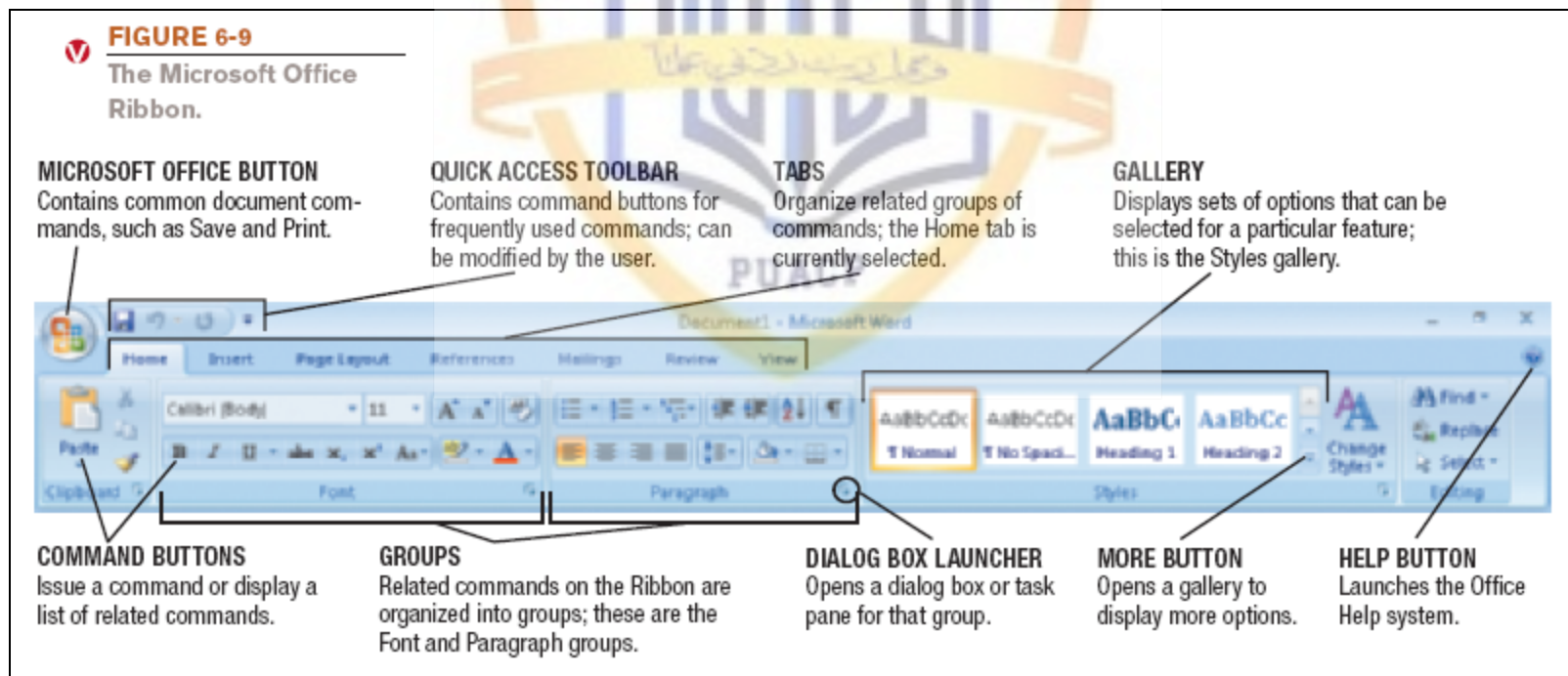
- Application programs today have a number of concepts and commands in common
 - Toolbars
 - Menus
 - Command buttons
 - Keyboard shortcuts
 - Ribbon
 - Mini toolbar

COMMAND	COMMAND BUTTON	KEYBOARD SHORTCUT	DESCRIPTION
Open		Ctrl+O	Opens a saved document from a storage medium, usually for editing or printing.
Save		Ctrl+S	Saves the current version of the document to a storage medium.
Print		Ctrl+P	Prints the current version of the document onto paper.
Cut		Ctrl+X	Moves the selected item to the clipboard.
Copy		Ctrl+C	Copies the selected item to the clipboard.
Paste		Ctrl+V	Pastes the contents of the clipboard to the current location.
Undo		Ctrl+Z	Undoes the last change to the document.
Close		Alt+F4	Closes the document. Any changes made to the document are lost if the document wasn't saved first.

FIGURE 6-8
Common application software commands.

The Ribbon

- Used in Microsoft Office 2007 and later
- Commands are organized into groups located on tabs
- Contextual: Commands appear as they are appropriate



Editing a Document

- Editing a document: Changing the content of the document, such as inserting or deleting words
 - Insertion point: Indicates the current location in a document, which is where the changes will be made
 - Typing text will insert it
 - Delete and Backspace key will delete text
 - Text and objects can typically be selected to be moved, copied, deleted, or formatted

Formatting a Document

- Formatting a document: Changing the appearance of the document
 - Changing font face, size, style, or color
 - Changing line spacing or margins
 - Adding page numbers or borders

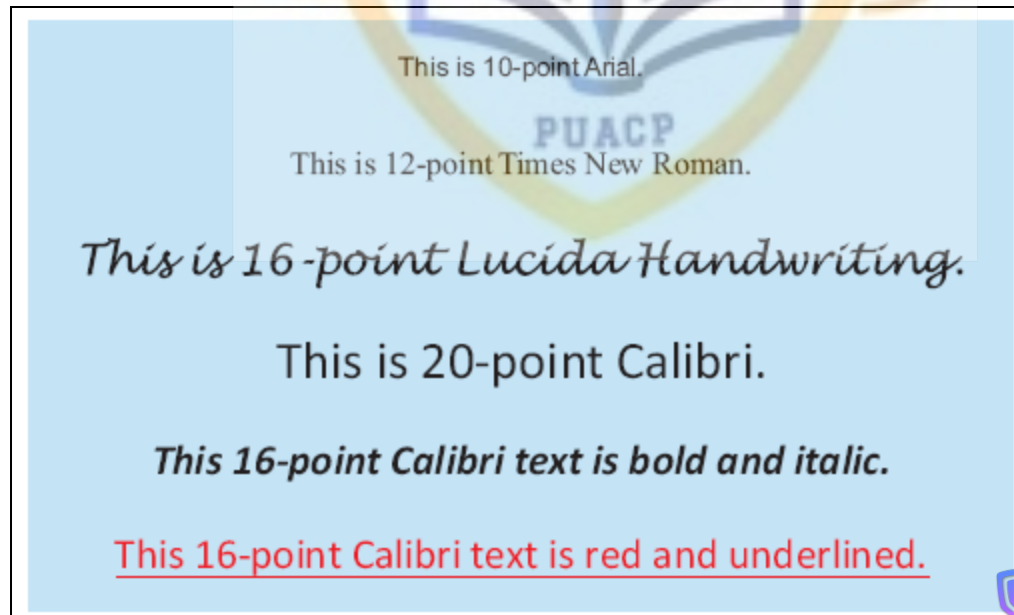
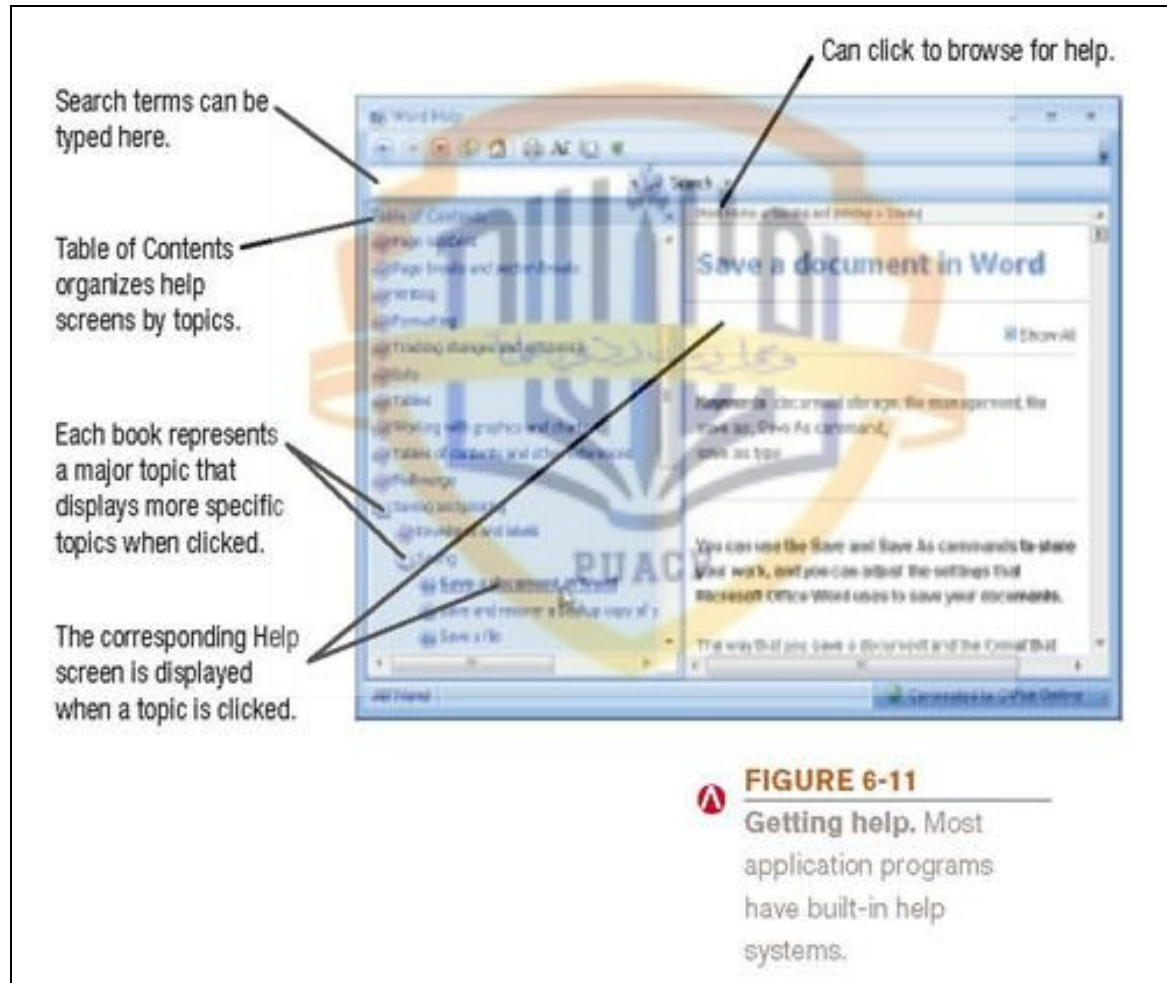


FIGURE 6-10
A **Fonts.** The font face, size, style, and color used with text can be specified in many applications programs.

Getting Help

- Help for software programs
 - Often built into the program
 - Table of Contents
 - Browsing
 - Search
 - Online help (via manufacturer's Web site and independent sites)
 - Offline help (periodicals, books, tutorial videos, classes)

Getting Help



Quick Quiz

1. Software programs that are distributed on the honor system and can be legally and ethically shared with others to try out the software are referred to as

_____.

- a. shareware programs
- b. commercial software
- c. public domain software

2. True or False: Software purchased via the Internet is always in downloaded, not packaged, form.

3. A group of related software programs sold together as one unit is called a(n) _____.

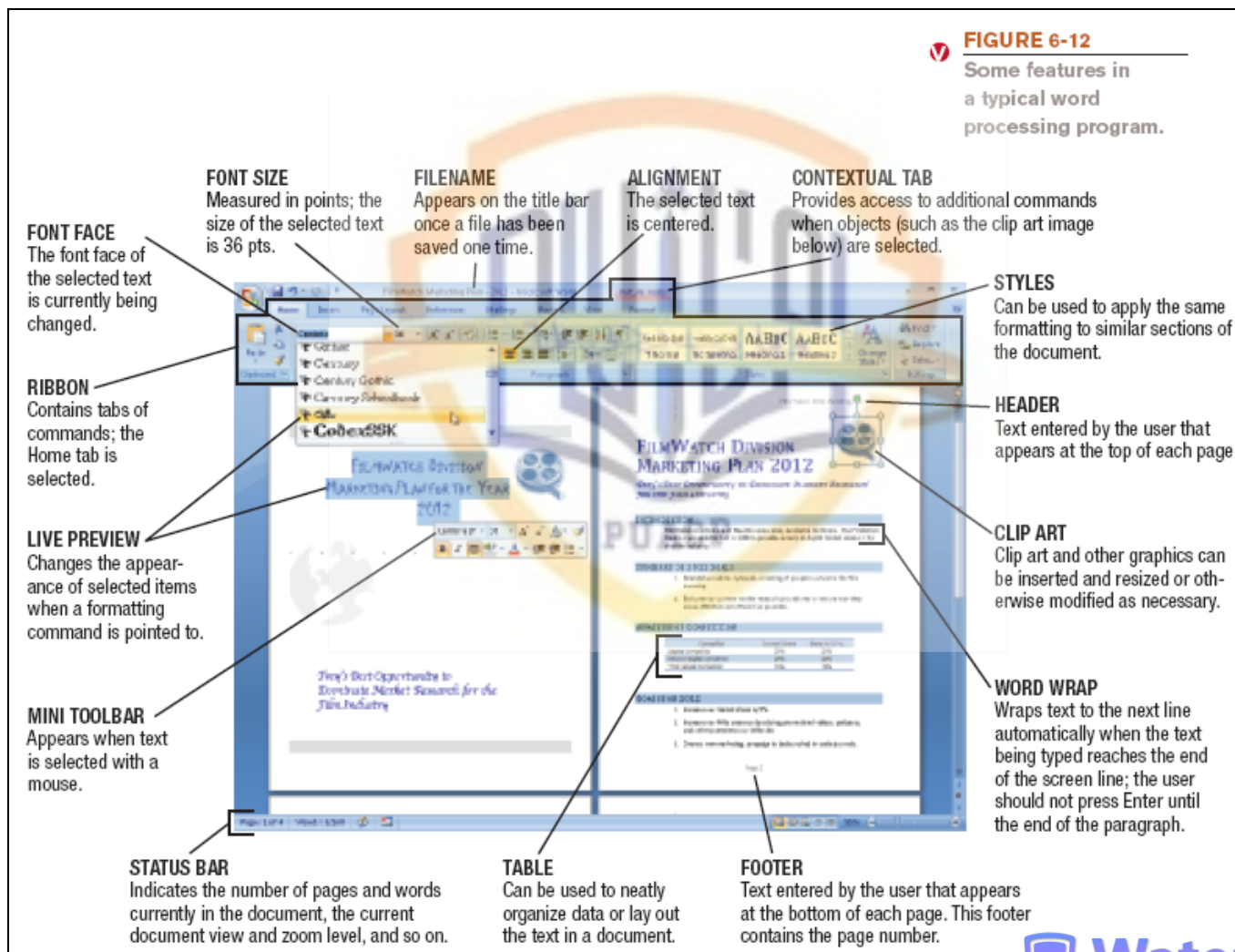
Answers:

1) a; 2) False; 3) software suite

Word Processing Concepts

- Word processing: Using a computer and word processing software to create, edit, save, and print written documents such as letters, contracts, and manuscripts
- Word processing software: Application software used to create, edit, save, and print written documents
- Common programs:
 - Microsoft Word
 - Corel WordPerfect
 - Apple Pages

Microsoft Word



Creating a Word Processing Document

- Word wrap: Automatically returns the insertion point to the next line when the end of the screen line is reached
- Character formatting (font face, size, style, or color)
- Paragraph formatting (line spacing, indentation, tabs, alignment, and styles)
- Page formatting (margins, paper size, orientation, headers, footers, etc)
- Document formatting (footnotes, end notes, table of contents, index, background, theme)

Tables, Graphics, and Templates

- Tables: Allow content to be organized in a table consisting of rows and columns
- Graphics or drawing features: Allow images to be inserted into a document (clip art, photographs, drawn images, etc.) and then modified
- Templates: Help users create new documents quickly



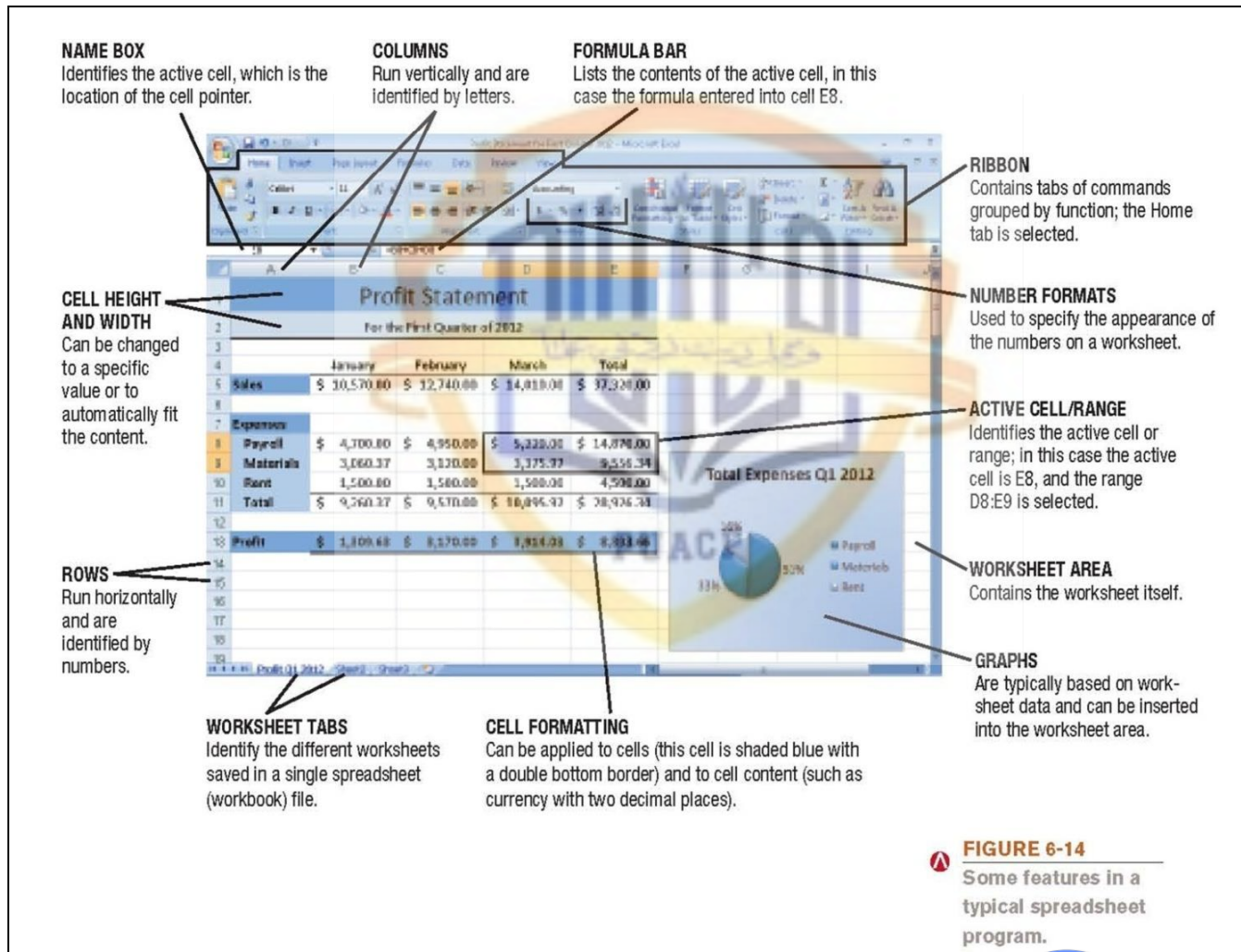
Word Processing on the Web

- Most word processing programs today include Web-related features allowing you to:
 - Send a document as an e-mail message
 - Include Web page hyperlinks in documents
 - Create or modify Web pages
 - Create and publish blogs
 - Collaborate with others online

Spreadsheet Concepts

- Spreadsheet software: Used to create electronic spreadsheets
- Spreadsheets: Documents containing a group of numbers and other data organized into rows and columns
- Worksheet: A single spreadsheet
- Workbook: A collection of worksheets saved in a single file
- Most widely used spreadsheet programs:
 - Microsoft Excel
 - Corel Quattro Pro
 - Apple Numbers

Microsoft Excel



Creating a Spreadsheet

- Worksheets are divided into rows and columns
- Cell: The intersection of a row and a column
 - Each cell is identified by a cell address
 - Cell pointer marks current cell
 - Groups of cells are called ranges or blocks; can be selected with mouse or keyboard
 - Data is entered into current cell

Entering Data into a Spreadsheet Cell

- Cells can contain:
 - Labels: Text-based entry in a worksheet cell that identifies data on the worksheet
 - Constant values: Numerical entry in a worksheet cell
 - Formulas: Performs mathematical operations on the content of other cells
 - Usually reference cell address, not the current data in a cell
 - Use mathematical operators + - / * ^
 - e.g. =B8+C8+D8
 - Functions: Named programmed formula
 - e.g. @SUM(B8..D8)

Entering Data into a Spreadsheet Cell

FIGURE 6-15
Universal
mathematical
operators.

SYMBOL	OPERATION
+	Addition
-	Subtraction
*	Multiplication
/	Division
^	Exponentiation

EXAMPLES OF FUNCTIONS

= SUM (range)	Calculates the sum of all values in a range.
= MAX (range)	Finds the highest value in a range.
= MIN (range)	Finds the lowest value in a range.
= AVERAGE (range)	Calculates the average of values in a range.
= FV (rate, number of payments, payment amount)	Calculates the future value of an annuity at a specified interest rate.
= PMT (rate, number of payments, loan amount)	Calculates the periodic payment for a loan.
= IF (conditional expression, value if true, value if false)	Supplies the values to be displayed if the conditional expression is true or if it is false.
= NOW ()	Inserts the current date and time.

FIGURE 6-16
Common
spreadsheet
functions.

Absolute vs. Relative Cell Referencing

- Relative cell references: Cell addresses are adjusted as the formula is copied
 - The default in most spreadsheet programs
 - e.g. B6
- Absolute cell references: Cell addresses are not adjusted as the formula is copied
 - Used when the formulas should always use the value in a particular cell, regardless of in which cell the formula is placed, such as for:
 - Constant value (sales tax rate, etc.)
 - Percent computation (always divide by total)
 - e.g. \$B\$6

Absolute vs. Relative Cell References

FIGURE 6-17
Relative vs. absolute
cell referencing.

COPYING WITH RELATIVE CELL REFERENCES

In most formulas, cell addresses are relative and will be adjusted as the formula is copied.

	A	B	C	D
1		Cones	Sundaes	Total
2	April	600	200	800
3	May	800	500	1300
4	June	1500	600	2100
5	Total			
6				

Formula in cell D2: $=B2+C2$

Formula in cell D4: $=B4+C4$

Results when the formula in cell D2 is copied to cells D3 and D4.

COPYING WITH ABSOLUTE CELL REFERENCES

A dollar sign (\$) marks a cell reference as absolute; it will be copied exactly as it appears in the source cell.

	A	B	C	D
1		Cones	Sundaes	Total
2	April	600	200	800
3	May	800	500	800
4	June	1500	600	800
5	Total			
6				

Formula in cell D2: $=\$B\$2+\$C\2

Results when the formula in cell D2 is copied to cells D3 and D4.

Formula in cell D4: $=\$B\$2+\$C\2

IMPROPER USE

	A	B	C	D	E
1		Cones	Sundaes	Total	Percent
2	April	600	200	800	19.05%
3	May	800	500	1300	30.95%
4	June	1500	600	2100	50.00%
5	Total			4200	100.00%
6					

Formula in cell E2: $=D2/\$D\5

Results when the formula in cell E2 is copied to cells E3 and E4.

PROPER USE

Formula in cell E4: $=D4/\$D\5

Charts and What-If Analysis

- Most spreadsheet programs include some type of charting or graphing capability
 - Can create charts from the data in the spreadsheet (don't have to reenter it)
 - Charts change accordingly if the data in the spreadsheet changes
- What-if analysis (sensitivity analysis)
 - Allows you to change certain numbers on a spreadsheet and immediately see the results of that change
 - Tool frequently used to help make business decisions

Spreadsheets and the Web

- Most spreadsheet programs have built-in Web capabilities, such as to:
- Save the current worksheet as a Web page
- Insert hyperlinks inserted into worksheet cells
- Select and copy ranges of cells to a Web publishing or word processing program to insert spreadsheet data into a Web page as a table
- Collaborate online

Quick Quiz

1. Changing the line spacing of a document is an example of which word processing feature?
 - a. word wrap
 - b. editing
 - c. formatting
2. True or False: A label is a special type of named formula, such as SUM to add up a group of cell values.
3. A spreadsheet document created in a spreadsheet program is often called a(n)
_____.

Answers:

1) c; 2) False; 3) worksheet

Database Concepts

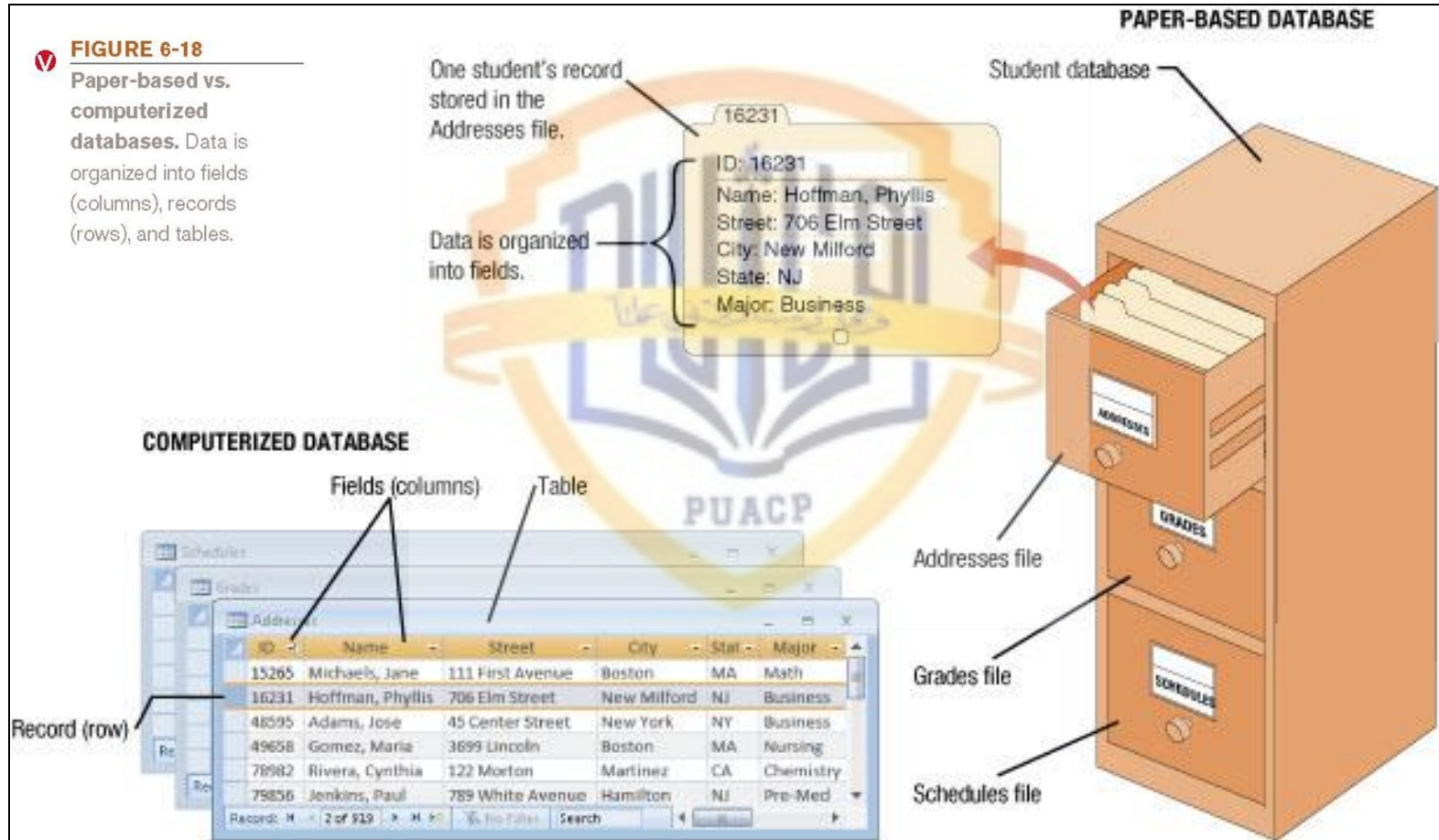
- Database: A collection of related data that is stored in a manner enabling information to be retrieved as needed
- Database management system (DBMS): Software that allows the creation and manipulation of an electronic database
- Most widely used database programs
 - Microsoft Access
 - Oracle Database
 - IBM DB2

Database Concepts

- Most PC-based databases are relational databases and organized into fields (columns), records (rows), and tables
- Field (column): A single type of data to be stored in a database
- Record (row): A collection of related fields
- Table: collection of related records
- Database file: Collection of related tables

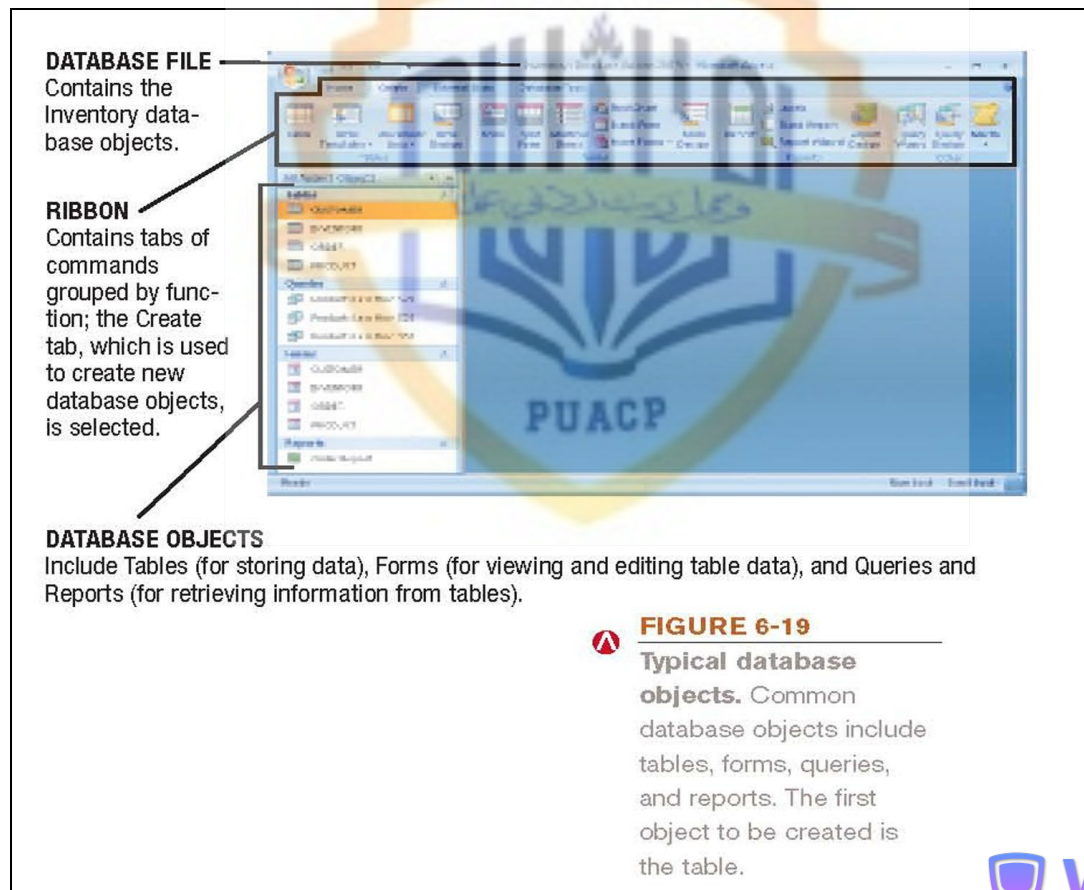
Database Concepts

FIGURE 6-18
Paper-based vs. computerized databases. Data is organized into fields (columns), records (rows), and tables.



Microsoft Access

- Databases contain a variety of objects (such as tables, queries, forms, reports)

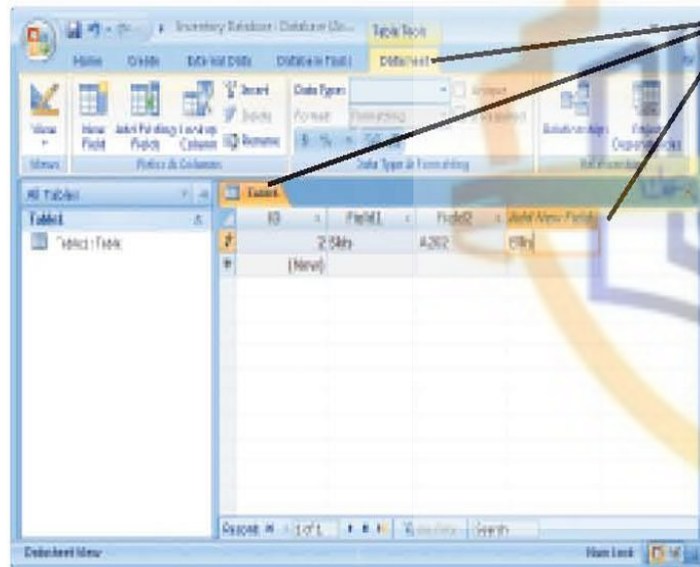


Creating a Database

- Database file is created first
- Tables can then be created
- Typically, the table structure is specified first
 - Includes a list of fields and their properties, including:
 - Field name (unique identifying name)
 - Data type (text, number, date, object)
 - Field size (maximum number of characters)
 - Default value (initial content of the field)
 - The table is named and saved
 - Tables can be created in either Datasheet or Design view

Creating a Database

DATASHEET VIEW



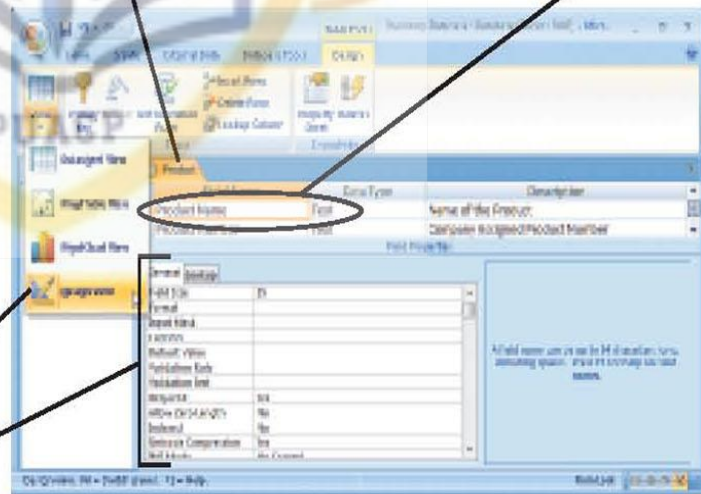
As data is entered into a new table using Datasheet view, the structure of the table is created.

The table has been saved as "Product" and is in Design view.

FIGURE 6-20

Creating a database table.

This field was renamed "Product Name" and is a Text field.



Click to select Design View.

Properties of the Product Name field.

DESIGN VIEW

Creating a Database

- After the table structure has been created, data is entered into the table
 - Datasheet view: Lists all data and looks similar to a spreadsheet
 - Form: Created by the user and typically displays just one record at a time
 - In either case, data is entered into the database table
- Either Datasheet view or a form can be used to view, modify, or delete data

Creating a Database

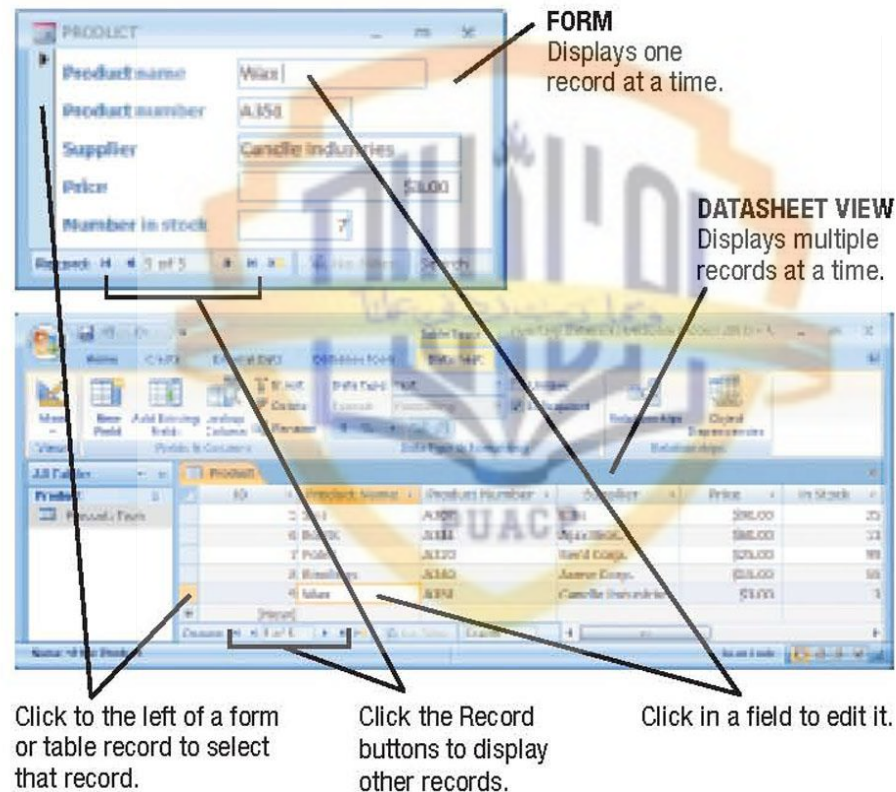


FIGURE 6-21

Table data can be modified using a form or the Datasheet view.



Queries and Reports

- Query: A question; a request for specific information from the database
 - Contains criteria to specify the records and fields to be included in the query results
 - Is named and saved so it can be run again at a later time
 - Results always include the current database data
- Reports: Created when a more formal output is desired
 - Associated with either a table or a query
 - Data is displayed in the specified format

Queries

QUERY DESIGN SCREEN

This query will display only the records that meet the specified criteria each time the query is retrieved.

Only these three fields will be displayed in the query results.

Field	Product Name	Product Number	Price
Table:	PRODUCT	PRODUCT	PRODUCT
Sort:	Ascending		
Show:	☒	☒	☒
Criteria:			<25

Only the records in which the price is less than \$25 will be displayed in the query results.

Product Name	Product Number	Price
Bindings	A240	\$15.00
Wax	A351	\$3.00

Record: 1 of 2 | No Filter | Search

QUERY RESULTS

The two records meeting the specified criteria are displayed.

FIGURE 6-22
Creating and using a database query.

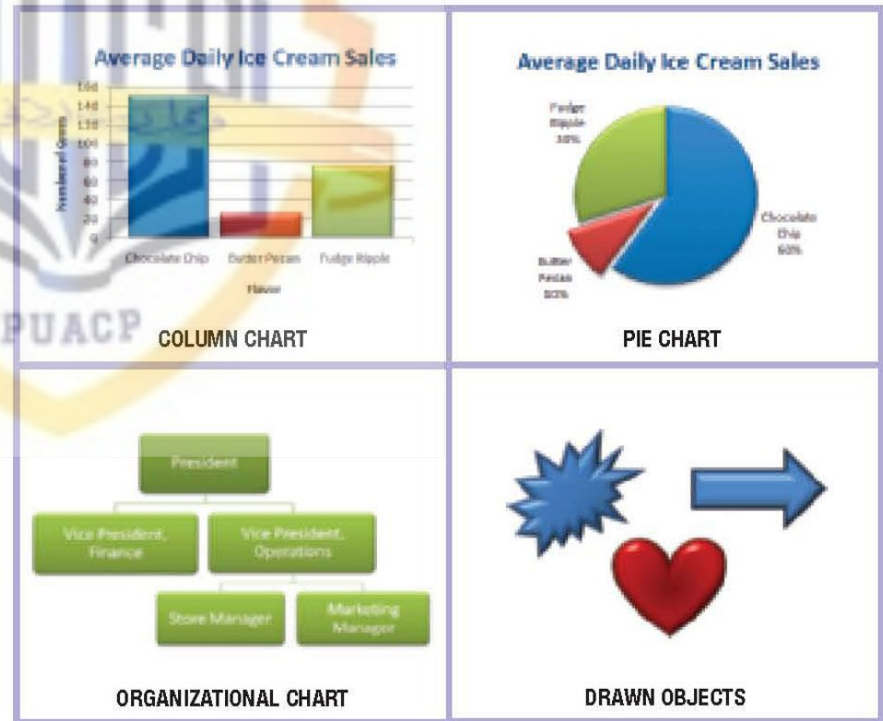
Databases and the Web

- Many Web sites use one or more databases to:
 - Keep track of inventory
 - Allow searching for people, documents, products, or other information
 - Place real-time orders
- Anytime you type keywords in a search site or hunt for a product on a retail store's Web site using its search feature, you are using a Web database

Presentation Graphic Concepts

- Presentation graphic:
An image designed to visually enhance a presentation
 - Can be used in electronic slide shows, as well as in printed reports
- Slide: A one-page presentation graphic that can be displayed in a group with others to form an electronic slide show

FIGURE 6-23
Examples of presentation graphics.



Presentation Graphic Concepts

- Electronic slide show: A group of electronic slides that are displayed one after the other on a computer monitor or other display device
- Presentation graphics software: Used to create presentation graphics
- Most common presentation graphics programs are:
 - Microsoft PowerPoint
 - Corel Presentations
 - Apple Keynote

Microsoft PowerPoint



FIGURE 6-24

Some features in a typical presentation graphics program.

NEW SLIDES

Use the Home tab to add a new slide.

SLIDES TAB

Click to see thumbnails of the slides in the presentation.

OUTLINE TAB

Click to see the titles and main text contained on the slides.

TEXT

Can be formatted using commands on the Ribbon or the Mini toolbar, like in other Office programs.

RIBBON

Contains tabs of commands grouped by function; the Insert tab is selected.

CLIP ART AND OTHER GRAPHICS

Can be inserted and resized or otherwise modified.

SLIDES

Can contain elements such as text, clip art, photographs, bulleted lists, charts, and video clips. Often a theme, color scheme, or design layout is used to apply a universal appearance to all slides.

DRAWN OBJECTS

Can be added, formatted, and resized; include lines, shapes, and more.

VIDEO CLIPS

Can be inserted into slides and set up to play automatically when the slide is loaded or when the video placeholder is clicked.

STATUS BAR

Includes information such as the current slide number and the total number of slides.

SOUND CLIPS

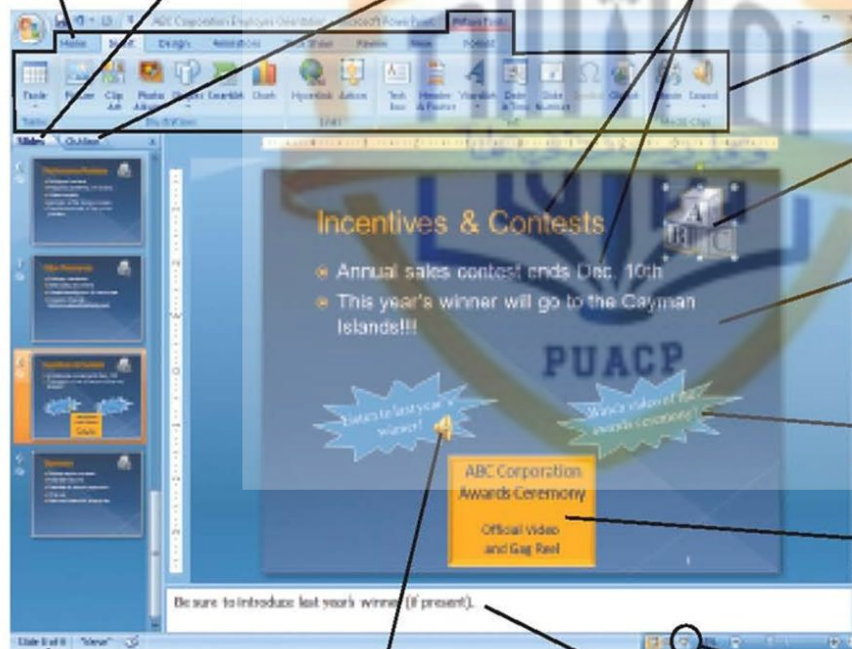
Can be inserted into slides and set up to play automatically when the slide is loaded or when the sound placeholder is clicked.

NOTES

Can be added so they can be seen by the presenter during the slide show or included on printouts.

SLIDE SHOW BUTTON

Click to run the slide show beginning with the current slide.



Creating a Presentation

- Templates and preformatted slide layouts can often be used
- New slides can be added as needed
- Slides can contain a variety of elements
 - Text
 - Images
 - Charts
 - Audio clips
 - Video clips
- Slides can be formatted as needed
- Multimedia and animation can be added to slides
 - Animation
 - Transitions



Finishing a Presentation

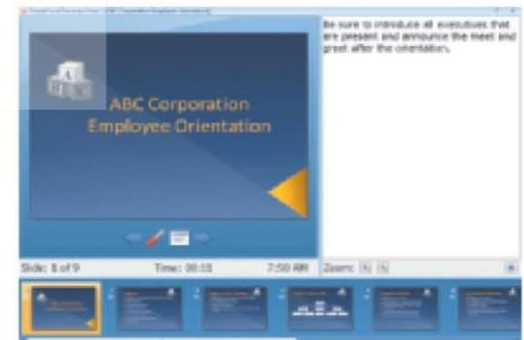
- Slide sorter view: Used to rearrange the slide order
- Show can be set up to run automatically or on mouse clicks
- Can add speaker notes, recorded narration, etc.
- Can print the slides to create overhead transparencies or an audience handout
- Speaker tools:
 - Speaker notes and pens
 - Presenter view

FIGURE 6-25
Running an electronic slide show.



SLIDE SHOW VIEW

Displays the slide show for the audience in full screen with the software interface hidden. Slides can be advanced at predetermined intervals or by clicking the mouse or pressing the spacebar.



PRESENTER VIEW

Seen only by the presenter on a different display device; includes thumbnails, a timer, speaker notes, annotation tools, and so forth.

Presentation Graphics and the Web

- Presentation graphics programs can be used to generate Web pages or Web page content
- Slides can include hyperlinks
- Users can usually control Web-based presentations accessed via a Web browser



Graphics and Multimedia Concepts

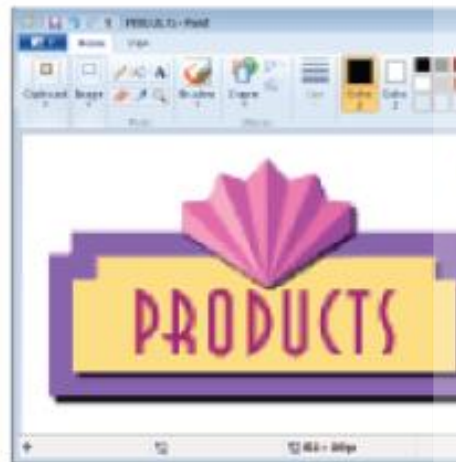
- Graphics: Graphical images, such as digital photographs, clip art, scanned drawings, and original images created using a software program
- Multimedia: Typically refers to any application that contains more than one type of media
 - Often used to refer to audio or video content
- Large variety of graphics and multimedia software, such as to:
 - Create or modify graphics
 - Edit digital audio or video
 - Play multimedia files
 - Burn CDs and DVDs

Graphics Software

- Graphics software: Used to create or modify images
 - Painting programs: Typically used to create bit-mapped images (Microsoft Paint)
 - Drawing programs: Typically create images using mathematical formulas (Adobe Illustrator, CorelDRAW)
 - Image editing or photo editing programs: Designed for touching up or modifying images
 - Correcting brightness/contrast
 - Cropping/eliminating red eye
 - Optimize file size for the Web
 - Adobe Photoshop, Apple iPhoto, etc.

Graphics Software

FIGURE 6-26
Graphics software.



PAINTING PROGRAMS

Typically create images pixel by pixel so images cannot be layered or resized.



DRAWING PROGRAMS

Typically create images using mathematical formulas so images can consist of multiple objects that can be layered, and the images can be resized without distortion.

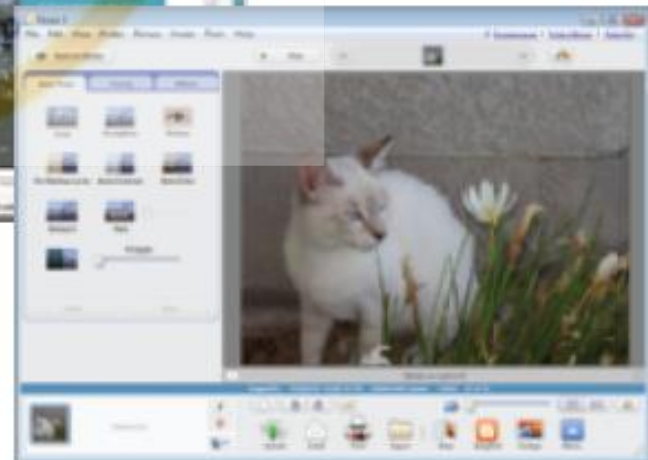


PHOTO EDITING PROGRAMS

Allow users to edit digital photos.

Audio Editing Software

- Audio editing software: Used to create and edit audio files (narrations, podcasts, etc.)
 - Capture sound from a microphone
 - Capture sound from a CD using ripping software
 - Edit and apply special effects
 - Windows Sound Recorder, Apple Garage Band, etc.



 **FIGURE 6-27**
Audio editing
software.

Video Editing and DVD Authoring Software

- Video editing software: Used to modify existing videos
 - Prepare video clips for presentations or Web sites
- DVD authoring software: Used to create DVD content
 - Importing videos, creating menus, burning onto a DVD
- DVD burning software: Used to record data on recordable or rewritable DVDs
 - Roxio Creator, Apple iMovie/iDVD, etc.

Video Editing Software

FIGURE 6-28

Video creation software. Often includes both video editing and DVD authoring capabilities.

VIDEO EDITING

Allows users to import and edit video; the finished video can be stored in a file or included in a DVD presentation.

Click to burn the finished presentation to a DVD.

The DVD includes the video clips and uses the titles and appearance designated by the user.



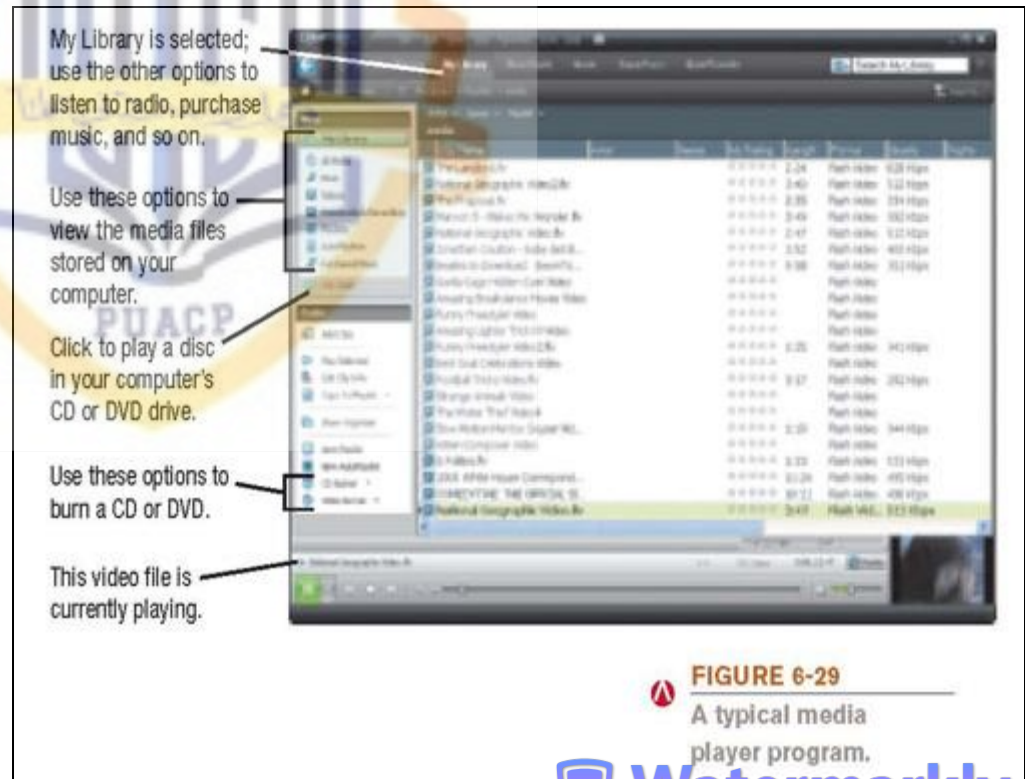
Video clips can be edited as needed; the timeline is used to crop out sections of the current video clip.

DVD AUTHORING

Allows users to import and organize photos, video, and music into a finished DVD presentation.

Media Players

- Media players: Programs designed to play audio and video files
 - Music CDs, downloaded music, online audio
 - Downloaded and online video
 - RealPlayer, Apple QuickTime, Windows Media Player, etc.



Graphics, Multimedia, and the Web

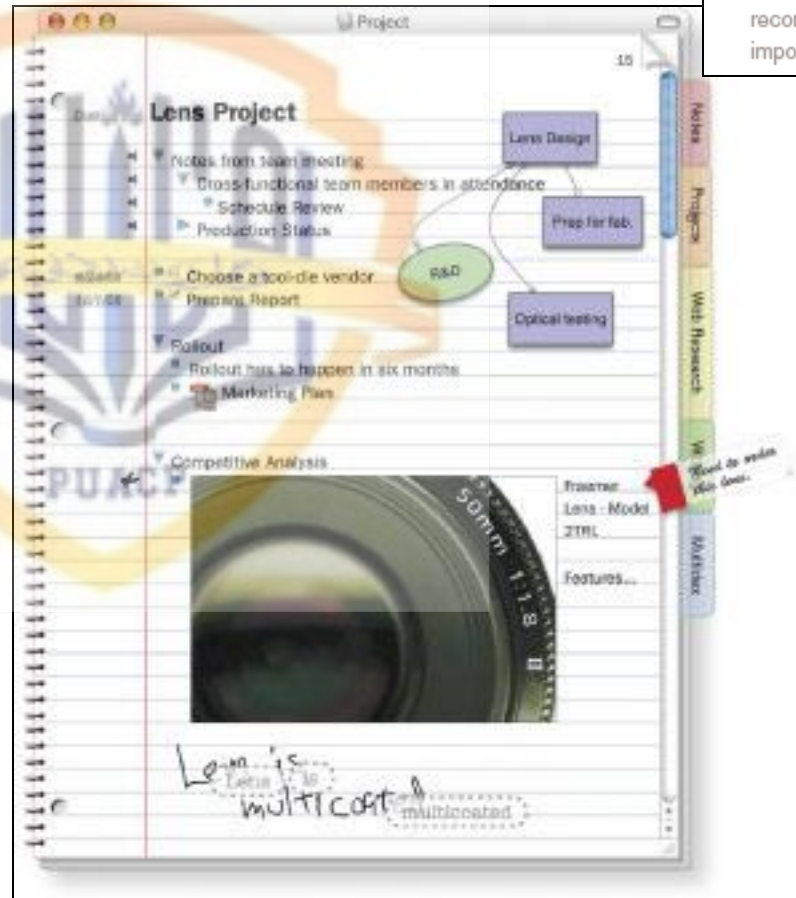
- Graphics and multimedia software is often used by individuals and businesses alike to create Web sites or content to be shared via the Web
 - Games
 - Tutorials
 - Videos
 - Demonstrations
 - Other content
- Creating multimedia Web sites is discussed in Chapter 10



Other Types of Application Software

- Desktop and personal publishing software
- Educational software
- Entertainment software
- Reference software
- Note taking software and Web notebooks

FIGURE 6-31
Note taking software.
Allows individuals to record and organize important data.



Other Types of Application Software

- CAD and other types of design software
- Accounting and personal finance software
- Project management, collaboration, and remote access software



FIGURE 6-33
File sending applications. This Web-based application allows you to easily send large files to others via the Internet.



FIGURE 6-32
Accounting software. This iPhone application helps individuals keep track of business expenses.

Quick Quiz

1. A single type of data to be included in a database (viewed as a column in a table) is called a

_____.

- a. record
- b. field
- c. table

2. True or False: Each slide in a slide presentation can contain only one type of element, such as text, an image, or a video clip.

3. Recording content on a CD or DVD disc is referred to as _____ the disc.

Answers:

1) b; 2) False; 3) burning

Summary

- The Basics of Application Software
- Word Processing Concepts
- Spreadsheet Concepts
- Database Concepts
- Presentation Graphics Concepts
- Graphics and Multimedia Concepts
- Other Types of Application Software